

Foundations of Arithmetic Power Geometry X: Spectral Entropy Theory and Energy Control on Compactified Modular Curves

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Abstract

This paper develops the tenth volume of Arithmetic Power Geometry (APG) by formulating a spectral entropy theory for APG deformation energy on compactified modular curves. Earlier APG volumes introduced local entropy-governed closure defects, information-geometric stabilization, integrated defect functionals, scale-corrected invariants, discrete prime-chain regularization, conditional APG–Arakelov height coupling, discrepancy-source compatibility, the Spectral–Green Bridge, and source concentration theory. The present paper proves the entropy–spectral energy theorem

$$E_{\text{APG}} \leq C_2 \lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2,$$

where E_{APG} is the APG Dirichlet energy, $H(W)$ is the Shannon entropy of the APG Euclidean weight distribution, and $\lambda_1(\Delta_{\text{APG}})$ is the first positive eigenvalue of the APG Laplacian. The theorem is a derived bridge result: it follows by combining the APG VIII Spectral–Green Bridge with the APG IX Source Concentration Theorem. The paper also introduces the spectral entropy ratio

$$R_{\text{spec}} = \frac{H(W)^2}{\lambda_1(\Delta_{\text{APG}})},$$

which linearizes the main inequality as $E_{\text{APG}} \leq C_2 R_{\text{spec}}$. The principal contribution of APG X is therefore not a new proof of spectral-gap stability, but a clean synthesis showing that APG energy is controlled by entropy once the spectral gap is fixed. This isolates spectral-gap stability and conductor/modular-degree dominance as the remaining tasks in the APG spectral-conductor program. The paper does not claim a proof of Fermat’s Last Theorem, the abc conjecture, the Szpiro conjecture, the Birch and Swinnerton-Dyer conjecture, Open Problem 15.1, or the full APG–Arakelov Projection Theorem.

Keywords: Arithmetic Power Geometry; spectral entropy; Shannon entropy; Laplacian; spectral gap; modular curves; Arakelov geometry; Dirichlet energy; source concentration; spectral conductor program.

1 Introduction

Arithmetic Power Geometry (APG) is a deformation-theoretic framework for studying algebraic closure relations under exponent variation. Instead of fixing an exponent and varying only coordinate fields, APG treats the exponent as a deformation parameter over a localized power-deformation space. This perspective was introduced in APG I, where the leading local closure-defect term near the Euclidean target was shown to be governed by Shannon entropy (Akhtar, 2026a; Shannon, 1948; Cover and Thomas, 2006). APG II extended the framework using information-geometric stabilization and coordinate stability methods inspired by modern information geometry (Akhtar, 2026b; Amari, 2016; Nielsen, 2020). APG III introduced the integrated closure-defect functional, APG IV studied coordinate-dependent damping and scale-corrected defect structures, and APG V replaced ill-defined continuous conductor paths with discrete prime-chain regularization (Akhtar, 2026c,d,e).

The later APG program shifted toward the APG–Arakelov architecture. APG VI formulated a conditional height-coupling program using Poisson energy and modular parametrization. APG VII established discrepancy-source compatibility and reduced the projection problem to boundary estimates involving cusp widths, vertical components, and Green-energy distortion fields (Akhtar, 2026f,g). APG VIII proved the Spectral–Green Bridge, converting the Green-energy obstruction into a precise spectral-source estimate on compactified modular curves (Akhtar, 2026h). APG IX then proved the source-concentration side of the program by bounding the mean-zero APG source in L^2 by entropy (Akhtar, 2026i).

The purpose of APG X is to turn these two results into a standalone spectral entropy theory. The main theorem states that APG energy satisfies

$$E_{\text{APG}} \leq C_2 \lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2. \quad (1)$$

This theorem is mathematically simple but structurally important. It says that the APG energy problem has been reduced to two factors: entropy and the spectral gap. Since $H(W)$ is already controlled by APG source theory, the remaining analytic obstruction is the possible collapse of $\lambda_1(\Delta_{\text{APG}})$. Thus APG X is a bridge milestone in the five-paper route toward the spectral-conductor program:

$$\text{APG IX} \longrightarrow \text{APG X} \longrightarrow \text{APG XI} \longrightarrow \text{APG XII} \longrightarrow \text{APG XIII}.$$

APG IX proves source control; APG X converts source control into energy control; APG XI must prove spectral-gap stability; APG XII must prove conductor/modular-degree dominance; and APG XIII would combine those estimates to prove the final spectral-conductor inequality.

The manuscript is deliberately scoped. Equation (1) is not presented as a new independent spectral theorem. It is a derived theorem whose proof depends on APG VIII and APG IX. This distinction is essential for mathematical clarity. It prevents the entropy-energy bridge from being confused with the still-open spectral-gap estimate required in APG XI.

2 Logical Dependency Structure

The development of APG X depends on the following chain:

$$\text{APG I} \implies \text{Entropy Defect Theory}$$

$$\text{APG IX} \implies \text{Source Concentration}$$

$$\text{APG VIII} \implies \text{Spectral-Green Bridge}$$

$$\text{APG VIII} + \text{APG IX} \implies \text{APG X}.$$

Consequently APG X introduces no additional source theorem and no additional spectral theorem. Its role is to establish the entropy–energy bridge.

3 Classical and APG Background

3.1 Classical setting

The APG spectral-conductor program is situated within classical arithmetic geometry, the theory of modular curves, Arakelov intersection theory, spectral geometry, and the modularity framework surrounding Fermat’s Last Theorem. Algebraic and arithmetic geometry provide the structural language of varieties, divisors, line bundles, heights, and rational points ([Hartshorne, 1977](#); [Lang, 1983, 2002](#); [Hindry and Silverman, 2000](#); [Bombieri and Gubler, 2006](#); [Silverman, 1986, 2007](#)). Arakelov theory enriches arithmetic intersection theory by including archimedean metric data ([Arakelov, 1974](#); [Faltings, 1984](#); [Soule, 1992](#); [Gillet and Soule, 1990](#); [Zhang, 1993](#)). Modular curves and modular parametrizations form a central bridge between arithmetic geometry and automorphic forms ([Shimura, 1971](#); [Deligne, 1971](#); [Diamond and Shurman, 2005](#); [Katz and Mazur, 1985](#)).

The proof of Fermat’s Last Theorem proceeds through the Frey curve, Serre’s modularity philosophy, Ribet’s level-lowering theorem, and Wiles–Taylor–Wiles modularity lifting ([Frey, 1986](#); [Serre, 1987](#); [Ribet, 1990](#); [Diamond, 1996](#); [Taylor and Wiles, 1995](#); [Wiles, 1995](#); [Darmon et al., 1997](#)). APG does not replace this classical machinery. Its aim is different: it builds an exponent-deformation language that can measure closure defects, source distributions, and deformation energy while respecting the boundaries of known arithmetic theory.

The broader analytic and arithmetic context also includes classical modular-function theory, analytic number theory, Diophantine approximation, and the geometry of modular curves ([Hecke, 1928](#); [Mazur, 1977](#); [Iwaniec and Kowalski, 2004](#); [Vojta, 1987](#); [Abbes and Ullmo, 1997](#)).

3.2 The APG development from I to IX

APG I proved the local entropy-governed closure-defect expansion at the Euclidean target ([Akhtar, 2026a](#)). APG II formulated a stabilized information-geometric layer and clarified coordinate

stability (Akhtar, 2026b). APG III introduced the integrated closure-defect functional A_0 and connected it to a conditional Frey-curve height-obstruction program (Akhtar, 2026c). APG IV identified the failure of universal damping and motivated scale-corrected invariants (Akhtar, 2026d). APG V corrected continuous conductor language by replacing it with discrete prime-chain regularization (Akhtar, 2026e). APG VI introduced Poisson energy and the modular-efficiency invariant in a conditional APG–Arakelov height-coupling setting (Akhtar, 2026f). APG VII proved discrepancy-source compatibility and reduced the projection problem to explicit boundary estimates (Akhtar, 2026g). APG VIII proved a Spectral–Green Bridge, and APG IX proved entropy-controlled source concentration (Akhtar, 2026h,i).

APG 0 collects this architecture, records its established theorem base, and explicitly distinguishes proved internal results from conditional projection programs and open arithmetic claims (Akhtar, 2026j). This paper follows that discipline: it proves only the entropy–spectral energy bridge and states all remaining arithmetic and spectral tasks as open.

4 Foundational Definitions

Let

$$X = X_0(N)^*(\mathbb{C})$$

be the compactified modular curve equipped with a smooth APG-compatible metric g_{APG} . Let $d\mu$ be the associated volume form and set

$$V = \text{Vol}(X) = \int_X d\mu.$$

Let ρ_{APG} be a nonnegative APG source density normalized by

$$\int_X \rho_{\text{APG}} d\mu = 1. \tag{2}$$

Define the mean-zero source fluctuation

$$F_{\text{APG}} = \rho_{\text{APG}} - \frac{1}{V}. \tag{3}$$

Then

$$\int_X F_{\text{APG}} d\mu = 0.$$

The positive APG Laplacian is defined by

$$\Delta_{\text{APG}} = -\text{div}_{\text{APG}} \nabla_{\text{APG}}. \tag{4}$$

Since X is compact and the metric is smooth, Δ_{APG} is a nonnegative self-adjoint elliptic operator on $L^2(X)$. Its spectrum is discrete:

$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots.$$

Because X is connected, the zero eigenspace consists of constants, and therefore

$$\lambda_1(\Delta_{\text{APG}}) > 0.$$

This is standard elliptic spectral theory on compact Riemannian manifolds and compact Riemann surfaces (Buser, 1992; Chavel, 1984; Jost, 2017; Rudin, 1976; Sarnak, 1995).

The APG Poisson potential u_{APG} is the unique mean-zero solution of

$$\Delta_{\text{APG}} u_{\text{APG}} = F_{\text{APG}}, \quad \int_X u_{\text{APG}} d\mu = 0. \quad (5)$$

The associated APG Dirichlet energy is

$$E_{\text{APG}} = \int_X |\nabla_{\text{APG}} u_{\text{APG}}|^2 d\mu. \quad (6)$$

5 Entropy Architecture

Let $(a, b) \in (\mathbb{R}_{>0})^2$. The APG Euclidean weight vector is

$$W = (w_a, w_b), \quad w_a = \frac{a^2}{a^2 + b^2}, \quad w_b = \frac{b^2}{a^2 + b^2}. \quad (7)$$

The Shannon entropy of W is

$$H(W) = -w_a \log w_a - w_b \log w_b. \quad (8)$$

For the binary APG weight distribution,

$$0 \leq H(W) \leq \log 2.$$

APG I proved that the local closure-defect near the Euclidean exponent $n = 2$ has leading entropy coefficient. In the notation of APG I,

$$\delta_K = \frac{H(W)}{2} |K_P| + O(K_P^2), \quad K_P = n - 2, \quad K_P \rightarrow 0. \quad (9)$$

This is the first point at which entropy becomes an APG deformation invariant.

APG IX strengthens the use of entropy at the source level. Instead of merely describing local sensitivity, entropy controls the size of the mean-zero source field in L^2 . This shift is decisive for the spectral theory because the Poisson problem responds to L^2 -source size.

6 The APG VIII Spectral–Green Bridge

The APG VIII theorem gives the analytic bridge from source fluctuation to Dirichlet energy. In the notation of this paper, it states

$$E_{\text{APG}} \leq \lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2. \quad (10)$$

For completeness, we recall the proof structure. Expanding F_{APG} in an orthonormal eigenbasis $\{\phi_j\}_{j \geq 1}$ of mean-zero eigenfunctions,

$$F_{\text{APG}} = \sum_{j \geq 1} c_j \phi_j, \quad \Delta_{\text{APG}} \phi_j = \lambda_j \phi_j.$$

The mean-zero Poisson solution is

$$u_{\text{APG}} = \sum_{j \geq 1} \frac{c_j}{\lambda_j} \phi_j.$$

Hence

$$E_{\text{APG}} = \int_X u_{\text{APG}} \Delta_{\text{APG}} u_{\text{APG}} d\mu = \sum_{j \geq 1} \frac{|c_j|^2}{\lambda_j} \leq \lambda_1^{-1} \sum_{j \geq 1} |c_j|^2 = \lambda_1^{-1} \|F_{\text{APG}}\|_2^2.$$

This proves (10). The theorem is purely analytic once the compact APG metric is fixed.

7 The APG IX Source Concentration Theorem

APG IX proves that the APG source fluctuation is entropy-controlled. In the notation of this paper, the theorem states

$$\rho_{\text{APG}} \in L^2(X), \quad \|\rho_{\text{APG}}\|_2 < \infty, \quad (11)$$

and

$$\left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2 \leq C_1 H(W)^2. \quad (12)$$

The proof in APG IX uses the decomposition

$$\rho_{\text{APG}} - \frac{1}{V} = \alpha_{\text{APG}} \sigma, \quad \|\sigma\|_2 = 1, \quad (13)$$

combined with an entropy-amplitude estimate

$$\alpha_{\text{APG}}^2 \leq C_1 H(W)^2. \quad (14)$$

Then

$$\left\| \rho_{\text{APG}} - \frac{1}{V} \right\|_2^2 = \|\alpha_{\text{APG}} \sigma\|_2^2 = \alpha_{\text{APG}}^2 \|\sigma\|_2^2 = \alpha_{\text{APG}}^2 \leq C_1 H(W)^2.$$

APG IX also handles cusp neighborhoods by using a cusp-regularized APG metric whose volume density contains an exponential damping factor. Under polynomial-logarithmic cusp growth, the corresponding cusp contribution is L^2 -integrable.

8 Main Result: Spectral Entropy Theory

Theorem 8.1 (Entropy–Spectral Energy Bridge Theorem). *Assume the APG VIII Spectral–Green Bridge (10) and the APG IX Source Concentration Theorem (12). Then there exists a structural constant $C_2 > 0$ such that*

$$E_{\text{APG}} \leq C_2 \lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2. \quad (15)$$

One may take $C_2 = C_1$, where C_1 is the APG IX source-concentration constant.

Proof. By APG VIII,

$$E_{\text{APG}} \leq \lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_2^2.$$

By APG IX,

$$\left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_2^2 \leq C_1 H(W)^2.$$

Substitution gives

$$E_{\text{APG}} \leq \lambda_1(\Delta_{\text{APG}})^{-1} C_1 H(W)^2.$$

Setting $C_2 = C_1$ proves

$$E_{\text{APG}} \leq C_2 \lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2.$$

□

Corollary 8.2 (Entropy Determines Energy Scale). *Assume $\lambda_1(\Delta_{\text{APG}}) \geq c > 0$.*

Then

$$E_{\text{APG}} = O(H(W)^2).$$

Proof. By Theorem 8.1 and the assumption $\lambda_1(\Delta_{\text{APG}}) \geq c > 0$, we have

$$E_{\text{APG}} \leq C_2 c^{-1} H(W)^2.$$

Therefore $E_{\text{APG}} = O(H(W)^2)$.

□

Hence entropy determines the asymptotic energy scale of APG deformation provided the spectral gap remains uniformly bounded below.

Remark 8.3 (Status of the theorem). Theorem 8.1 is a derived theorem, not an independent proof of spectral-gap stability. Its role is to combine the APG VIII and APG IX estimates into a single entropy–spectral energy statement.

9 Spectral Entropy Invariants

The main theorem suggests two related spectral entropy quantities.

Definition 9.1 (Multiplicative spectral entropy). Define

$$S_{\text{spec}} = H(W) \lambda_1(\Delta_{\text{APG}}). \quad (16)$$

This quantity measures the product of entropy and the spectral gap. However, it does not linearize the APG X energy inequality. Indeed, since

$$\lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2 = \frac{H(W)^3}{S_{\text{spec}}},$$

small S_{spec} corresponds to large possible energy. Thus S_{spec} is diagnostically useful but not the cleanest energy-control invariant.

Definition 9.2 (Spectral entropy ratio). Define

$$R_{\text{spec}} = \frac{H(W)^2}{\lambda_1(\Delta_{\text{APG}})}. \quad (17)$$

The following reformulation is immediate.

Corollary 9.3 (Linear spectral entropy control). *Under the hypotheses of Theorem 8.1,*

$$E_{\text{APG}} \leq C_2 R_{\text{spec}}. \quad (18)$$

Proof. This follows directly from (15) and the definition of R_{spec} . $\square$

10 Energy Scaling Consequences

Corollary 10.1 (Entropy collapse implies energy collapse under a stable gap). *Suppose that $\lambda_1(\Delta_{\text{APG}}) \geq c > 0$ along a family of APG systems. If $H(W) \rightarrow 0$, then $E_{\text{APG}} \rightarrow 0$.*

Proof. By Theorem 8.1,

$$E_{\text{APG}} \leq C_2 c^{-1} H(W)^2.$$

The right-hand side tends to zero as $H(W) \rightarrow 0$. $\square$

Corollary 10.2 (Spectral instability criterion). *Assume $H(W) \leq H_0$. If E_{APG} is large, then $\lambda_1(\Delta_{\text{APG}})$ must be small in the sense that*

$$\lambda_1(\Delta_{\text{APG}}) \leq \frac{C_2 H_0^2}{E_{\text{APG}}}. \quad (19)$$

Proof. From Theorem 8.1,

$$E_{\text{APG}} \leq C_2 \lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2 \leq C_2 \lambda_1(\Delta_{\text{APG}})^{-1} H_0^2.$$

Rearranging gives (19) whenever $E_{\text{APG}} > 0$. $\square$

Corollary 10.3 (Uniform entropy bound). *Since $H(W) \leq \log 2$, one always has*

$$E_{\text{APG}} \leq C_2 (\log 2)^2 \lambda_1(\Delta_{\text{APG}})^{-1}. \quad (20)$$

$$H(W) \implies \left\| \rho_{\text{APG}} - \frac{1}{V} \right\|_2^2 \implies E_{\text{APG}} \implies \text{Spectral-Conductor Program.}$$

11 Reduction of the Spectral-Conductor Program

The APG spectral-conductor target is

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_2^2 \leq C \deg(\pi) \log N. \quad (21)$$

APG IX reduces the source term to entropy. APG X packages this into the spectral entropy ratio.

Proposition 11.1 (APG X reduction principle). *Assume APG IX. To prove (21), it is sufficient to prove*

$$\lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2 \leq C_{ast} \deg(\pi) \log N \quad (22)$$

for a structural constant $C_{ast} > 0$.

Proof. By APG IX,

$$\left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_2^2 \leq C_1 H(W)^2.$$

Therefore

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_2^2 \leq C_1 \lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2.$$

If (22) holds, then

$$C_1 \lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2 \leq C_1 C_{ast} \deg(\pi) \log N.$$

Setting $C = C_1 C_{ast}$ proves (21). $\square$

Remark 11.2. Proposition 11.1 does not establish the conductor estimate (22). It merely reduces Open Problem 15.1 to proving (22). The proof of (22) is deferred to APG XII.

12 Role in the Five-Paper Route

The five-paper route to the spectral-conductor theorem may be summarized as follows.

Volume	Role	Principal Target
APG IX	Source concentration	$\ \rho_{\text{APG}} - 1/V\ _2^2 \leq C_1 H(W)^2$
APG X	Spectral entropy bridge	$E_{\text{APG}} \leq C_2 \lambda_1^{-1} H(W)^2$
APG XI	Spectral gap stability	$\lambda_1(\Delta_{\text{APG}}) \geq c/(\log N)^A$
APG XII	Conductor dominance	$H(W)^2 (\log N)^A \leq C_3 \deg(\pi) \log N$
APG XIII	Final synthesis	Spectral-conductor theorem

APG X completes its assigned role in this route. It does not complete the whole route. The decisive remaining analytic step is APG XI, because without a lower bound for $\lambda_1(\Delta_{\text{APG}})$, the inverse spectral factor may still be too large.

13 Numerical Illustration

The following model calculation illustrates the theorem. Suppose

$$H(W) = 0.60, \quad \lambda_1(\Delta_{\text{APG}}) = 0.25, \quad C_2 = 1.$$

Then

$$R_{\text{spec}} = \frac{H(W)^2}{\lambda_1(\Delta_{\text{APG}})} = \frac{0.36}{0.25} = 1.44.$$

Therefore APG X gives

$$E_{\text{APG}} \leq 1.44.$$

If instead the spectral gap collapses to 0.05 while entropy remains 0.60, then

$$R_{\text{spec}} = \frac{0.36}{0.05} = 7.20,$$

and the energy bound becomes much weaker. This illustrates why spectral-gap stability is the core task of APG XI.

14 Discussion

APG X marks the point at which entropy and spectral geometry become a single analytic mechanism. APG I showed that entropy governs local deformation sensitivity. APG IX showed that entropy controls source fluctuation. APG VIII showed that source fluctuation controls energy through the spectral gap. APG X combines these facts and states the resulting structure as a unified spectral entropy theory.

The paper also clarifies what has not been proved. It does not prove a uniform spectral gap. It does not prove conductor-scale dominance. It does not prove the APG–Arakelov Projection Theorem. Its value lies in isolating the remaining problem precisely:

$$\lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2 \leq C \deg(\pi) \log N.$$

This is the correct bridge between source concentration and conductor-scale arithmetic growth.

15 Scope and Non-Claims

The following statements are not claimed in this paper:

- (i) APG X does not prove Fermat’s Last Theorem.
- (ii) APG X does not prove the abc conjecture.
- (iii) APG X does not prove the Szpiro conjecture.
- (iv) APG X does not prove the Birch and Swinnerton-Dyer conjecture.
- (v) APG X does not prove the full APG–Arakelov Projection Theorem.

- (vi) APG X does not prove the spectral-gap estimate required in APG XI.
- (vii) APG X does not prove the conductor/modular-degree dominance estimate required in APG XII.
- (viii) APG X does not prove Open Problem 15.1 by itself.

The paper proves exactly the entropy–spectral energy bridge stated in Theorem 8.1, conditional only on the established APG VIII and APG IX inputs.

16 Open Problems

Open Problem 16.1 (APG XI spectral-gap stability). Prove that the APG Laplacian satisfies

$$\lambda_1(\Delta_{\text{APG}}) \geq \frac{c}{(\log N)^A}$$

for constants $c > 0$ and $A \geq 0$. The strongest possible version would be a uniform lower bound $\lambda_1(\Delta_{\text{APG}}) \geq c > 0$.

Open Problem 16.2 (Metric comparison). Prove a comparison between the APG Laplacian and a standard Arakelov or hyperbolic Laplacian:

$$C^{-1}\Delta_{\text{std}} \leq \Delta_{\text{APG}} \leq C\Delta_{\text{std}}.$$

Such a comparison would provide a route to spectral-gap transfer.

Open Problem 16.3 (APG XII conductor dominance). Prove

$$H(W)^2(\log N)^A \leq C_3 \deg(\pi) \log N.$$

This would show that the arithmetic conductor/modular-degree side dominates the spectral entropy energy side.

Open Problem 16.4 (APG XIII final synthesis). Combine APG IX, APG XI, and APG XII to prove

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_2^2 \leq C \deg(\pi) \log N.$$

17 Conclusion

This paper establishes APG X as the spectral entropy bridge of the APG program. By combining the APG VIII Spectral–Green Bridge with the APG IX Source Concentration Theorem, we prove

$$E_{\text{APG}} \leq C_2 \lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2.$$

Equivalently, the spectral entropy ratio

$$R_{\text{spec}} = \frac{H(W)^2}{\lambda_1(\Delta_{\text{APG}})}$$

controls APG energy:

$$E_{\text{APG}} \leq C_2 R_{\text{spec}}.$$

This completes the assigned role of APG X in the five-paper route. The source cannot blow up independently of entropy, and APG energy is now controlled by entropy and the spectral gap. The remaining decisive task is APG XI: proving that the APG spectral gap does not collapse too rapidly with conductor level.

Final Status of APG X

APG X completes the entropy-to-energy component of the Arithmetic Power Geometry program.

The source-concentration problem is resolved by APG IX.

The energy-control problem is resolved by APG X.

The remaining unresolved tasks are:

1. APG XI: Spectral Gap Stability.
2. APG XII: Conductor and Modular Degree Dominance.
3. APG XIII: Final Spectral-Conductor Synthesis.

Accordingly APG X should be regarded as a completed bridge theorem rather than a final arithmetic theorem.

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